

Second-Moment Wedge Projection and the FVGQ Risk-Sharing Channel

A Standalone Research Report

Caught in the Crossfire Research Notes

May 1, 2026

Abstract

This report turns the informal argument that “a first-order BCA prototype projects second-moment uncertainty into first-order wedges” into a self-contained theoretical document. We work entirely within the Fernández-Villaverde–Guerrón-Quintana (2020; FVGQ) two-agent collateral RBC, whose three-stage timing, skin-in-the-game financial friction, and three stochastic-volatility shocks are reproduced in §3 so that the report can be read without recourse to the original paper. The central second-order result is a wedge projection equation: when the true data-generating process contains stochastic volatility, the investment wedge recovered by a Business Cycle Accounting (BCA) prototype must satisfy a forward-looking equation forced by the second-order risk correction in the true Euler equation; under AR(1) volatility, the projection admits a closed form, decomposable into precautionary, risk-premium, collateral-risk, and two-agent aggregation components. We then advance the projection to third order in the perturbation parameter (Theorem 3). The third-order extension introduces a vol-of-vol refinement of the linear coefficient, a state-dependent cross term $\Gamma(\hat{z}_t, \Omega_t)$, and an explicit asymmetric response of the BCA wedge to positive versus negative volatility innovations. These three new objects supply identifying handles that the second-order theory cannot deliver, including a structural explanation for the negative wide-grid Monte Carlo estimate of the collateral-risk sensitivity ν_θ . The report also contrasts the FVGQ collateral channel with a first-order working-capital channel, which projects into the after-tax labor wedge rather than the investment wedge.

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1 Purpose and Main Results

This report integrates the following six internal research notes:

- 2026-05-01-wedge-projection-derivation-skeleton.md
- 2026-05-01-fvgq-Phi-closed-form.md
- 2026-05-01-fvgq-Phi-numerical-validation.md
- 2026-05-01-fvgq-two-agent-risk-sharing-wedge.md
- 2026-05-01-fvgq-nu-theta-monte-carlo.md
- 2026-05-01-fvgq-aggregation-wedge-numerical.md

The common question is how a first-order BCA prototype absorbs the second-order uncertainty terms generated by a nonlinear DSGE model with stochastic volatility.

The main conclusions are as follows.

1. The BCA investment wedge satisfies the projection equation

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t),$$

where $\Phi(\Omega_t)$ is the second-order risk correction in the true Euler equation.

2. If Ω_t follows an AR(1) process, the projection has the closed form

$$\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1 - \omega_\tau)} + \frac{\phi}{2(1 - \omega_\tau \rho_\Omega)} (\Omega_t - \bar{\Omega}), \quad \phi = \Phi'(\bar{\Omega}).$$

Thus BCA can transform a second-moment stochastic-volatility signal into a persistent first-order investment wedge.

3. In an FVGQ-style financial-friction model, the risk correction can be decomposed as

$$\Phi^{\text{FVGQ}} = \Phi^{\text{prec}} + \Phi^{\text{rp}} + \Phi^{\text{coll}} + \Phi^{\text{agg}}.$$

These terms capture precautionary motives, risk premia, collateral risk, and the aggregation error induced by using total consumption in a two-agent economy.

4. Financial uncertainty in the FVGQ collateral mechanism is predicted to appear mainly in the investment wedge, with a positive sign under the baseline collateral-risk channel. The sign of macro uncertainty is not guaranteed, because the risk-premium covariance can dominate the precautionary component.
5. Numerical diagnostics from the FVGQ workspace qualify the simple closed-form comparison. The Monte Carlo estimate of the collateral-risk term ν_θ is small and negative in the current grid, while the two-agent aggregation correction is positive for both volatility channels. Aggregation therefore cannot explain the negative local-projection peak for the macro uncertainty shock.

6. A working-capital mechanism has a different empirical signature. If firms must finance wage payments in advance, the BCA after-tax labor wedge satisfies

$$\hat{\Lambda}_{\ell,t} = -\lambda_{wc}(q_t - \bar{q}) + O((q_t - \bar{q})^2), \quad \lambda_{wc} > 0.$$

Therefore a rise in the external finance spread lowers the after-tax labor wedge. This sign is first-order and does not depend on Hessian terms.

7. Extending the projection equation to third order in the perturbation parameter (Theorem 3) introduces three new objects: a refinement to the second-order coefficient via vol-of-vol amplification, a state-dependent cross term $\Gamma(\hat{z}_t, \Omega_t)$, and an asymmetric response of $\hat{\tau}_{x,t}$ to positive versus negative volatility innovations. The third-order extension is necessary to interpret the FVGQ generalized impulse response functions, because in second order alone the on-impact response to an uncertainty shock $u_{j,t}$ is identically zero; it is only at third order that $u_{j,t}$ enters the policy function.

2 Notation and Normalization

Hats denote log deviations from the deterministic steady state; for example, $\hat{C}_t = \log(C_t/\bar{C})$. The stochastic-volatility state is

$$\Omega_t = (\sigma_{A,t}^2, \sigma_{e,t}^2)', \quad \Delta\Omega_t = \Omega_t - \bar{\Omega}.$$

In the one-factor case, Ω_t can be treated as a scalar.

To avoid ambiguity in the labor-wedge sign convention, this report uses the after-tax labor wedge

$$\Lambda_{\ell,t} \equiv 1 - \tau_{\ell,t}^{\text{tax}}.$$

Thus $\hat{\Lambda}_{\ell,t} < 0$ means that the labor wedge deteriorates. If the same result is reported in terms of the labor-tax rate $\tau_{\ell,t}^{\text{tax}}$, the sign is reversed.

Remark 1 (Normalization of Φ). This report uses the convention

$$\hat{\tau}_{x,t} = \omega_{\tau} \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t).$$

Some of the underlying notes absorb the factor 1/2 into the definition of Φ . All formulas for ϕ_A , ϕ_e , $\kappa_{x,A}$, and $\kappa_{x,e}$ in this report use the convention above. A different convention rescales the coefficients but leaves the sign-identification logic unchanged.

3 The FVGQ 2020 Model Environment

The closed-form risk decomposition in §5, the two-agent aggregation correction in §6, and the numerical validation in §10 all take Fernández-Villaverde and Guerrón-Quintana (2020; henceforth FVGQ) as the reference data-generating process. The original paper presents the model in its Section 5. This section reproduces the equations, conventions, and calibration needed for the projection theory below, so that the report is self-contained.

3.1 Households, Three-Stage Timing, and Equity Claims

Time is discrete. There is a continuum of ex-ante identical households of measure one, each containing a continuum of members. Aggregate state is Markov, and a representative household head writes contingency plans for all members.

Three stages. Each period is divided into three stages.

1. *Decision stage.* Members are united and share assets, which consist of s_t units of an equity claim with price q_t . One claim entitles the holder to one unit of capital next period. The household head splits assets across members and writes contingent allocation plans.
2. *Production stage.* Each member draws a shock determining her type. With probability π the member becomes an entrepreneur; with probability $1 - \pi$ she becomes a worker. Entrepreneurs invest in capital but do not work; workers supply labor but cannot invest. Both can trade equity claims, subject to the financial frictions specified below.
3. *Consumption-investment stage.* Workers receive wage income; entrepreneurs collect capital rental income, depreciate their equity, and invest i_t units of the consumption good through a one-to-one technology yielding i_t units of new capital. Consumption takes place. The period ends; member identities are reshuffled.

The fraction π is calibrated to $\pi = 0.06$, matching the share of firms that invest in any given quarter in U.S. Compustat data.

3.2 Production Technology and Capital Utilization

A representative competitive firm produces a final good using capital services k_t^D and labor ℓ_t :

$$y_t = A_t (k_t^D)^\alpha \ell_t^{1-\alpha}, \quad k_t^D = u_t k_t, \quad (\text{Y-FVGQ})$$

where u_t is the capital utilization rate. The utilization rate controls depreciation through the reduced-form schedule

$$\delta(u_t) = \delta_0 + \delta_1(u_t - 1) + \frac{\delta_2}{2}(u_t - 1)^2, \quad (\text{Dep-FVGQ})$$

with δ_0 the deterministic SS depreciation rate. A higher u_t accelerates depreciation, providing the firm with an internal margin to absorb shocks without changing k_t .

TFP follows a level AR(1) with stochastic volatility:

$$A_t = (1 - \rho_A)\bar{A} + \rho_A A_{t-1} + e^{\sigma_{A,t}} \varepsilon_{A,t}, \quad \varepsilon_{A,t} \sim \mathcal{N}(0, 1), \quad \bar{A} = 1. \quad (\text{A-FVGQ})$$

The reflecting barrier $\bar{A} > 0$ guarantees positivity but is irrelevant for local approximations. The conditional volatility $\sigma_{A,t}$ is itself a stochastic process described in §3.4 below.

The firm's first-order conditions equate marginal products to factor prices:

$$\alpha \frac{y_t}{k_t^D} = r_t, \quad (1 - \alpha) \frac{y_t}{\ell_t} = w_t.$$

3.3 Skin-in-the-Game Financial Friction

The defining feature of FVGQ is a Kiyotaki-Moore-style collateral constraint on the entrepreneur's portfolio. Two parameters control how much of the entrepreneur's investment can be financed externally.

- For *new capital*: the entrepreneur can sell at most a fraction $\theta_t \in [0, 1]$ of new equity to outside investors, retaining at least $(1 - \theta_t)i_t$. The pledgeability θ_t is time-varying.
- For *existing capital* $(1 - \delta(u_t))s_t$: the entrepreneur can sell at most a fraction $\phi \in [0, 1]$ to outside investors, retaining $(1 - \phi)(1 - \delta(u_t))s_t$. The resaleability ϕ is constant.

Combining the two restrictions delivers the skin-in-the-game lower bound on entrepreneur equity holdings:

$$s_{t+1}^E \geq (1 - \theta_t)i_t + (1 - \phi)(1 - \delta(u_t))s_t. \quad (\text{F-FVGQ})$$

The analogous bound for workers, $s_{t+1}^W \geq (1 - \phi)(1 - \delta(u_t))s_t$, never binds in equilibrium because workers are net buyers of equity.

This formulation differs from environments with idiosyncratic productivity risk (Bernanke-Gertler-Gilchrist, Buera-Quadrini-Shin) in that the friction scales with the size of investment rather than with productivity heterogeneity. It also differs from Jermann and Quadrini (2012) in that θ_t is exogenous and time-varying; this exogeneity is what supplies the second source of stochastic volatility.

The constraint (F-FVGQ) binds in equilibrium whenever the price of capital exceeds one ($q_t > 1$), which is the relevant case throughout the calibration.

3.4 Three Uncertainty Shocks (Stochastic Volatility)

The model has three structural level shocks — TFP A_t , preference shifter d_t , and pledgeability θ_t — each driven by an innovation with time-varying conditional volatility:

$$A_t = (1 - \rho_A)\bar{A} + \rho_A A_{t-1} + e^{\sigma_{A,t}} \varepsilon_{A,t}, \quad (\text{Lvl-A})$$

$$d_t = (1 - \rho_d)\bar{d} + \rho_d d_{t-1} + e^{\sigma_{d,t}} \varepsilon_{d,t}, \quad (\text{Lvl-d})$$

$$\theta_t = (1 - \rho_\theta)\bar{\theta} + \rho_\theta \theta_{t-1} + e^{\sigma_{\theta,t}} \varepsilon_{\theta,t}. \quad (\text{Lvl-}\theta)$$

Each conditional volatility itself follows an AR(1):

$$\sigma_{j,t} = (1 - \rho_{\sigma_j})\sigma_j + \rho_{\sigma_j}\sigma_{j,t-1} + (1 - \rho_{\sigma_j}^2)^{1/2}\nu_j u_{j,t}, \quad u_{j,t} \sim \mathcal{N}(0, 1), \quad (\text{SV-}j)$$

for $j \in \{A, d, \theta\}$. The innovations $u_{j,t}$ are the *uncertainty shocks*: $u_{A,t}$ is the supply-side TFP uncertainty shock; $u_{d,t}$ is the demand-side preference uncertainty shock; and $u_{\theta,t}$ is the financial-friction uncertainty shock.

Notation reconciliation with the implementation. The remainder of this report (and the Dynare implementation in `fvgq2020_solveonly.mod`) writes the financial-friction volatility as $\sigma_{e,t}$ rather than $\sigma_{\theta,t}$, with associated innovation $u_{e,t}$ rather than $u_{\theta,t}$. The two notations refer to the same object. Sections 5–10 use $\sigma_{e,t}$ and $u_{e,t}$ to match the codebase.

3.5 Optimality Conditions and the Liquidity Wedge

The household head maximizes

$$\mathbb{E}_t \sum_{t=0}^{\infty} \beta^t d_t \left\{ \pi \frac{(c_t^E)^{1-\rho} - 1}{1-\rho} + (1-\pi) \frac{[c_t^W (1-\ell_t)^\eta]^{1-\rho} - 1}{1-\rho} \right\} \quad (\text{U-FVGQ})$$

subject to a household-level budget constraint and the financial constraint (F-FVGQ). Here ρ is the inverse intertemporal elasticity of substitution, $\eta > 0$ controls leisure curvature, and d_t is the demand shifter described in (Lvl-d).

After substituting the binding (F-FVGQ) into the household budget constraint, the modified budget constraint contains a positive premium on investment whenever $q_t > 1$. The Lagrange multiplier on this modified constraint defines a *liquidity wedge*:

$$\lambda_t = \frac{q_t - 1}{1 - \theta_t q_t}. \quad (\text{Liq-FVGQ})$$

Provided $1 < q_t < 1/\theta_t$, this wedge is positive: it prices the value of one unit of liquidity to the household head, who can convert it into $1/(1 - \theta_t q_t)$ units of capital via leveraged investment.

The household-head optimality conditions take the following form.

- **Worker labor supply.** Because workers face no borrowing constraint, the labor supply margin is undistorted by the financial friction:

$$\eta \frac{c_t^W}{1 - \ell_t} = w_t. \quad (\text{LS-FVGQ})$$

- **Intra-household consumption allocation.** Entrepreneurs face a liquidity premium, so their marginal utility exceeds that of workers:

$$(c_t^E)^{-\rho} = (1 + \lambda_t)(c_t^W)^{-\rho}(1 - \ell_t)^{\eta(1-\rho)}. \quad (\text{IH-FVGQ})$$

- **Investment Euler (capital pricing).** The price of equity claims satisfies

$$q_t = \mathbb{E}_t \left\{ \beta \frac{d_{t+1}}{d_t} \left(\frac{c_{t+1}^W}{c_t^W} \right)^{-\rho} \left(\frac{1 - \ell_{t+1}}{1 - \ell_t} \right)^{\eta(1-\rho)} \mathcal{P}_{t+1} \right\}, \quad (\text{IE-FVGQ})$$

where the per-claim payoff is

$$\mathcal{P}_{t+1} = r_{t+1}u_{t+1} + q_{t+1}(1 - \delta(u_{t+1})) + \pi\lambda_{t+1}[r_{t+1}u_{t+1} + q_{t+1}\phi(1 - \delta(u_{t+1}))]. \quad (\text{Pay-FVGQ})$$

The first two summands are the dividend and the depreciated capital gain. The $\pi\lambda_{t+1}$ term is the value of the liquidity services that the equity claim provides to entrepreneurs in the next period.

- **Capital utilization.** The marginal benefit of using capital more intensively equals the marginal cost from faster depreciation:

$$r_t = q_t \delta'(u_t). \quad (\text{Util-FVGQ})$$

- **Capital accumulation.**

$$k_{t+1} = (1 - \delta(u_t))k_t + \pi i_t. \quad (\text{Cap-FVGQ})$$

- **Goods market clearing.**

$$\pi c_t^E + (1 - \pi)c_t^W + \pi i_t = y_t. \quad (\text{GM-FVGQ})$$

Table 1: FVGQ 2020 calibration used throughout the report.

Parameter	Value	Target / Source
β : discount factor	0.994	exogenously chosen
ρ : relative risk aversion	2	exogenously chosen
π : fraction of entrepreneurs	0.06	investing-firm fraction $\simeq 0.24/\text{yr}$
η : leisure curvature	1.525	hours of work = 1/3
α : capital share	0.36	labor income share = 0.64
δ_0 : SS depreciation	0.0163	annual investment/capital = 0.065
δ_1 : linear utilization slope	0.0131	pinned by SS
δ_2 : utilization curvature	$0.33 \delta_1$	Comín and Gertler (2006)
ϕ : existing-capital resaleability	0.0954	capital/output = 3.30
$\bar{\theta}$: SS new-capital pledgeability	0.0954	set equal to ϕ
$\rho_A = \rho_d = \rho_\theta$: level persistences	0.95	Cooley and Prescott (1995)
$\rho_{\sigma_A} = \rho_{\sigma_d} = \rho_{\sigma_\theta}$: SV persistences	0.75	FVGQ et al. (2015a)
σ_A : SS TFP volatility	$\log(0.007)$	Cooley and Prescott (1995)
σ_d : SS preference volatility	$\log(0.032)$	FVGQ et al. (2015a)
σ_θ : SS friction volatility	$\log(0.008)$	Guerrón-Quintana and Jinnai (2019b)

3.6 Calibration

Table 1 reproduces the calibration used in `fvgq2020_solveonly.mod` and throughout the report. Two parameters (δ_1 and η) are pinned down by SS targets; all others are taken directly from the calibration in FVGQ Section 7.

The volatility innovation scale ν_j is calibrated so that a one-standard-deviation positive uncertainty innovation $u_{j,t} = 1$ doubles the level of $\sigma_{j,t}$ relative to its SS:

$$\nu_j = \frac{\log 2}{(1 - \rho_{\sigma_j}^2)^{1/2}}. \quad (\text{Nu-FVGQ})$$

With $\rho_{\sigma_j} = 0.75$ this gives $\nu_j \simeq 1.048$, which is the value used in the codebase.

The deterministic SS values that underpin §5 and §9 follow from this calibration. In particular, $\bar{R} = 1/\beta \simeq 1.0060$, $\bar{R}^k \simeq 0.0273$, and the BCA-Euler forward coefficient is

$$\omega_\tau = \frac{1 - \delta}{\bar{R}^k + 1 - \delta} \simeq 0.973.$$

Bibliographic note. References in this section — Comín and Gertler (2006), Cooley and Prescott (1995), Jermann and Quadrini (2012), Fernández-Villaverde et al. (2015a), and Guerrón-Quintana and Jinnai (2019b) — appear inline without a formal bibliography. A future revision will compile a `.bib` file. For the present draft they should be read as pointers to the standard literature listed in the FVGQ paper.

4 Generic BCA Wedge Projection

4.1 BCA Prototype Euler Equation

The representative-household Euler equation in a CKM-style BCA prototype can be written, after first-order log linearization around the deterministic steady state, as

$$\hat{\tau}_{x,t} - \gamma \tilde{C}_t = \mathbb{E}_t \left[-\gamma \tilde{C}_{t+1} + \omega_K \tilde{R}_{t+1}^k + \omega_\tau \hat{\tau}_{x,t+1} \right]. \quad (\text{BCA-E})$$

Here γ is relative risk aversion, $\tilde{C}_t$ and $\tilde{R}_{t+1}^k$ are the observed consumption and capital-return objects fed into the BCA prototype, and ω_K, ω_τ are steady-state coefficients.

Suppose that the true data-generating process contains stochastic volatility. Its Euler equation, expanded to second order, can be written as

$$-\gamma \tilde{C}_t = \mathbb{E}_t \left[-\gamma \tilde{C}_{t+1} + \omega_K \tilde{R}_{t+1}^k \right] + \frac{1}{2} \Phi(\Omega_t) + o(\|\Delta\Omega_t\|). \quad (\text{DGP-E})$$

The term $\Phi(\Omega_t)$ collects risk corrections generated by conditional variances, conditional covariances, and Hessian terms.

Feeding the data generated by (DGP-E) into the BCA prototype and comparing (BCA-E) with (DGP-E) gives the projection equation

$$\boxed{\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t)}. \quad (\text{P})$$

This equation is the key object of the report.

4.2 Closed-Form Projection

Assumption 1 (Stochastic-volatility state). The stochastic-volatility state satisfies

$$\Delta\Omega_{t+1} = R_\Omega \Delta\Omega_t + \eta_{t+1}, \quad \rho(R_\Omega) < 1,$$

where $\rho(R_\Omega)$ is the spectral radius.

Assumption 2 (Local differentiability). Φ is differentiable near $\bar{\Omega}$ and satisfies

$$\Phi(\Omega_t) = \Phi(\bar{\Omega}) + \phi' \Delta\Omega_t + o(\|\Delta\Omega_t\|).$$

Theorem 1 (Wedge Projection). *Under the assumptions above and $|\omega_\tau| < 1$, the locally bounded solution to the projection equation (P) is*

$$\boxed{\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1-\omega_\tau)} + \frac{1}{2} \phi' (I - \omega_\tau R_\Omega)^{-1} \Delta\Omega_t + o(\|\Delta\Omega_t\|)}. \quad (\text{T1})$$

If Ω_t is a scalar $AR(1)$ process, $\Delta\Omega_{t+1} = \rho_\Omega \Delta\Omega_t + \eta_{t+1}$, then

$$\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1-\omega_\tau)} + \underbrace{\frac{\phi}{2(1-\omega_\tau \rho_\Omega)}}_{\kappa_x} (\Omega_t - \bar{\Omega}) + o(|\Omega_t - \bar{\Omega}|). \quad (\text{T1s})$$

Proof. Forward iteration of (P) gives

$$\hat{\tau}_{x,t} = \frac{1}{2} \sum_{j=0}^{\infty} \omega_{\tau}^j \mathbb{E}_t \Phi(\Omega_{t+j}).$$

The local approximation implies

$$\mathbb{E}_t \Phi(\Omega_{t+j}) = \Phi(\bar{\Omega}) + \phi' R_{\Omega}^j \Delta \Omega_t + o(\|\Delta \Omega_t\|).$$

Substitution and the geometric-series identities

$$\sum_{j=0}^{\infty} \omega_{\tau}^j = \frac{1}{1 - \omega_{\tau}}, \quad \sum_{j=0}^{\infty} (\omega_{\tau} R_{\Omega})^j = (I - \omega_{\tau} R_{\Omega})^{-1}$$

yield (T1). The scalar formula is the one-dimensional special case. $\square$

The theorem has three immediate implications.

- **Persistence pass-through.** BCA may recover a wedge that looks like a persistent AR(1) first-order wedge, even though the underlying force is a stochastic-volatility state.
- **Amplification.** If $\omega_{\tau} \rho_{\Omega}$ is close to one, the second-order risk correction is amplified by $(1 - \omega_{\tau} \rho_{\Omega})^{-1}$.
- **Sign discipline.** The sign of the investment-wedge response is governed by $\phi = \Phi'(\bar{\Omega})$. This makes it possible to distinguish second-order Euler channels from first-order static-FOC channels.

5 FVGQ Closed Form for $\Phi(\Omega_t)$

5.1 Entrepreneur Euler Equation

This section specializes the generic projection equation (P) of §4 to the FVGQ environment described in §3. To keep the algebra transparent, we work in the representative-entrepreneur, always-binding-collateral, no-adjustment-cost limit of (F-FVGQ). The two-agent extension is restored in §6.

Let $\lambda_t = u'(c_t)$ be the entrepreneur's marginal utility of consumption. The bond and capital first-order conditions are

$$\lambda_t = \beta R_t \mathbb{E}_t \lambda_{t+1} + \mu_t, \tag{F1}$$

$$\lambda_t = \beta \mathbb{E}_t [\lambda_{t+1} (R_{t+1}^k + 1 - \delta)] + \mu_t \theta_t (1 - \delta). \tag{F2}$$

Eliminating the collateral multiplier μ_t gives

$$\lambda_t [1 - \theta_t (1 - \delta)] = \beta \mathbb{E}_t [\lambda_{t+1} (R_{t+1}^k + 1 - \delta)] - \beta R_t \theta_t (1 - \delta) \mathbb{E}_t \lambda_{t+1}. \tag{E-Euler}$$

This Euler equation is the source of the FVGQ risk-correction terms.

5.2 Risk-Correction Decomposition

A second-order expansion of (E-Euler), using $\hat{\lambda}_{t+1} = -\gamma\hat{c}_{t+1}$ under CRRA preferences, organizes the stochastic-volatility terms as

$$\Phi^{\text{FVGQ}}(\Omega_t) = \Phi^{\text{prec}}(\Omega_t) + \Phi^{\text{rp}}(\Omega_t) + \Phi^{\text{coll}}(\Omega_t). \quad (\text{FVGQ-Phi})$$

The three components are

$$\Phi^{\text{prec}}(\Omega_t) = \frac{1}{2}\gamma^2[1 - \bar{\theta}(1 - \delta)] \Sigma_{cc}(\Omega_t), \quad (\text{Prec})$$

$$\Phi^{\text{rp}}(\Omega_t) = -\gamma\beta(\bar{R}^k + 1 - \delta) \Sigma_{cR}(\Omega_t), \quad (\text{RP})$$

$$\Phi^{\text{coll}}(\Omega_t) = \beta\bar{R}\bar{\theta}(1 - \delta)\nu_\theta\sigma_{e,t}^2. \quad (\text{Coll})$$

Here

$$\Sigma_{cc}(\Omega_t) = \text{Var}_t(\hat{c}_{t+1}), \quad \Sigma_{cR}(\Omega_t) = \text{Cov}_t(\hat{c}_{t+1}, \hat{r}_{t+1}^k).$$

The interpretation is direct. The precautionary component is positive when uncertainty raises the conditional variance of future consumption. The risk-premium component is negative when consumption and capital returns are positively correlated. The collateral-risk component captures the effect of financial uncertainty on future borrowing capacity; if $\nu_\theta > 0$, this component raises the projected investment wedge.

5.3 Mapping to κ_x

Near the deterministic steady state, write

$$\Sigma_{cc}(\Omega_t) = a_{cc,A}\sigma_{A,t}^2 + a_{cc,e}\sigma_{e,t}^2, \quad (1)$$

$$\Sigma_{cR}(\Omega_t) = a_{cR,A}\sigma_{A,t}^2 + a_{cR,e}\sigma_{e,t}^2. \quad (2)$$

Then

$$\phi_A = \frac{1}{2}\gamma^2[1 - \bar{\theta}(1 - \delta)]a_{cc,A} - \gamma\beta(\bar{R}^k + 1 - \delta)a_{cR,A}, \quad (\text{phiA})$$

$$\phi_e = \frac{1}{2}\gamma^2[1 - \bar{\theta}(1 - \delta)]a_{cc,e} - \gamma\beta(\bar{R}^k + 1 - \delta)a_{cR,e} + \beta\bar{R}\bar{\theta}(1 - \delta)\nu_\theta. \quad (\text{phie})$$

Theorem 1 gives

$$\hat{\tau}_{x,t} = \kappa_{x,A}\Delta\sigma_{A,t}^2 + \kappa_{x,e}\Delta\sigma_{e,t}^2, \quad (\text{Kappa-map})$$

with

$$\kappa_{x,A} = \frac{\phi_A}{2(1 - \omega_\tau\rho_{\sigma_A})}, \quad \kappa_{x,e} = \frac{\phi_e}{2(1 - \omega_\tau\rho_{\sigma_e})}. \quad (\text{Kappa})$$

This mapping makes the FVGQ sign predictions testable. Financial uncertainty contains the collateral-risk term and is therefore expected to generate a positive investment-wedge projection in the baseline model. Macro uncertainty is ambiguous because the precautionary and risk-premium terms can have opposing signs.

6 Two-Agent Risk-Sharing Wedge

The full FVGQ model has workers and entrepreneurs. The BCA prototype, however, uses aggregate NIPA consumption C_t as the consumption of a representative agent. The marginal utility $u'(C_t)$ used by the BCA observer is therefore not the same as the entrepreneur's marginal utility $u'(c_{E,t})$, which is the object that prices investment in the true model. This difference creates an aggregation wedge.

6.1 Aggregation Identity

Let

$$C_t = \omega_W c_{W,t} + \omega_E c_{E,t}, \quad \omega_W = 0.94, \quad \omega_E = 0.06.$$

Define steady-state consumption shares

$$s_W = \frac{\omega_W \bar{c}_W}{\bar{C}}, \quad s_E = \frac{\omega_E \bar{c}_E}{\bar{C}}, \quad s_W + s_E = 1.$$

A second-order expansion of aggregate consumption gives

$$\hat{C}_t = s_W \hat{c}_{W,t} + s_E \hat{c}_{E,t} + \frac{1}{2} s_W s_E (\hat{c}_{E,t} - \hat{c}_{W,t})^2 + O(3). \quad (\text{Agg-C})$$

Let

$$\delta c_t \equiv \hat{c}_{E,t} - \hat{c}_{W,t}.$$

Under CRRA preferences,

$$\log u'(C_t) - \log u'(c_{E,t}) = -\gamma(\hat{C}_t - \hat{c}_{E,t}) = \gamma s_W \delta c_t - \frac{1}{2} \gamma s_W s_E \delta c_t^2 + O(3). \quad (\text{MU-gap})$$

The first term is a first-order risk-sharing gap between entrepreneurs and workers. The second term is a Jensen aggregation term.

6.2 Projection Contribution

Substituting the aggregation identity into the BCA Euler equation and subtracting the entrepreneur Euler equation yields

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \gamma s_W (\mathbb{E}_t \delta c_{t+1} - \delta c_t) + \frac{1}{2} \Phi^{\text{tot}}(\Omega_t). \quad (\text{P-2A})$$

The total risk correction is

$$\Phi^{\text{tot}} = \Phi^{\text{prec}} + \Phi^{\text{rp}} + \Phi^{\text{coll}} + \Phi^{\text{agg}}.$$

Under the normalization used in this report, the aggregation component is

$$\Phi^{\text{agg}}(\Omega_t) = \gamma s_W s_E \left[\gamma \text{Var}_t(\delta c_{t+1}) - \text{Cov}_t(\delta c_{t+1}, \hat{r}_{t+1}^k) \right]. \quad (\text{Agg-Phi})$$

If the factor 1/2 is instead absorbed into the definition of Φ , the right-hand side of (Agg-Phi) must be rescaled accordingly.

6.3 Linear Policy-Function Interpretation

Let the one-step-ahead innovation loading of consumption dispersion be

$$\delta c_{t+1} = \chi_A \varepsilon_{A,t+1} + \chi_\theta \varepsilon_{\theta,t+1} + \text{state terms},$$

and let the corresponding loading of capital returns be

$$\hat{r}_{t+1}^k = r_A \varepsilon_{A,t+1} + r_\theta \varepsilon_{\theta,t+1} + \text{state terms}.$$

If the two shocks are orthogonal, then

$$\text{Var}_t(\delta c_{t+1}) = \chi_A^2 \sigma_{A,t}^2 + \chi_\theta^2 \sigma_{e,t}^2, \quad (3)$$

$$\text{Cov}_t(\delta c_{t+1}, \hat{r}_{t+1}^k) = \chi_A r_A \sigma_{A,t}^2 + \chi_\theta r_\theta \sigma_{e,t}^2. \quad (4)$$

Hence the marginal aggregation coefficients are

$$\begin{aligned} \phi_A^{\text{agg}} &= \gamma s_W s_E \chi_A (\gamma \chi_A - r_A), & (\text{aggA}) \\ \phi_e^{\text{agg}} &= \gamma s_W s_E \chi_\theta (\gamma \chi_\theta - r_\theta). & (\text{agge}) \end{aligned}$$

Financial uncertainty typically makes entrepreneur consumption much more sensitive to collateral shocks than worker consumption. Therefore $\text{Var}_t(\delta c_{t+1})$ should rise with $\sigma_{e,t}^2$, reinforcing the positive investment-wedge projection. Macro uncertainty is less clear because both workers and entrepreneurs are exposed to TFP shocks, and the covariance term can offset the variance term.

7 Working-Capital Counter-Hypothesis

The FVGQ baseline is a second-order Euler-equation channel. A working-capital friction is different: it is a first-order static labor-demand channel. It should therefore project into the labor wedge rather than the investment wedge.

7.1 Exact Labor-Wedge Projection

Suppose firms must finance a fraction θ_{wc} of the wage bill before production. Let q_t denote the external finance spread. The firm's static condition is

$$A_t F_{L,t} = w_t [1 + \theta_{wc} q_t]. \quad (\text{WC-FOC})$$

The BCA intratemporal labor condition is

$$-\frac{U_L(C_t, L_t)}{U_C(C_t, L_t)} = \Lambda_{\ell,t}^{BCA} A_t F_{L,t}. \quad (\text{BCA-L})$$

The true household condition gives $-U_L/U_C = w_t$. Combining these conditions gives the exact projection identity

$$\Lambda_{\ell,t}^{BCA} = \frac{w_t}{A_t F_{L,t}} = \frac{1}{1 + \theta_{wc} q_t}. \quad (\text{WC-P})$$

Log-linearizing around $\bar{q}$ yields

$$\hat{\Lambda}_{\ell,t}^{BCA} = - \underbrace{\frac{\theta_{wc}}{1 + \theta_{wc} \bar{q}}}_{\lambda_{wc} > 0} (q_t - \bar{q}) + O((q_t - \bar{q})^2). \quad (\text{WC-LIN})$$

If the spread satisfies the first-order approximation

$$q_t - \bar{q} = \alpha_\xi \hat{\xi}_t + \alpha_\Omega (\Omega_t - \bar{\Omega}) + o(\|\hat{\xi}_t, \Omega_t - \bar{\Omega}\|), \quad \alpha_\xi > 0, \quad \alpha_\Omega \geq 0, \quad (\text{RP-WC})$$

then

$$\hat{\Lambda}_{\ell,t}^{BCA} = -\lambda_{wc} \alpha_\xi \hat{\xi}_t - \lambda_{wc} \alpha_\Omega (\Omega_t - \bar{\Omega}) + O(\hat{\xi}_t^2) + O((\Omega_t - \bar{\Omega})^2). \quad (\text{T2})$$

7.2 Economic Content

If a financial tightening $\hat{\xi}_t > 0$ or higher uncertainty raises the external finance spread, then

$$\hat{\Lambda}_{\ell,t}^{BCA} < 0.$$

This sign does not depend on Hessian terms, risk-premium covariance, or the numerical size of a simulation. Moreover, if $\hat{\xi}_t$ is a first-moment shock, the labor-wedge response is $O(\sigma)$, while a pure stochastic-volatility Hessian projection into the investment wedge is $O(\sigma^2)$. The two mechanisms are therefore distinguishable by both wedge location and local order.

8 Unified Sign Identification

Theorem 2 (Wedge Sign Identification). *Consider two competing mechanisms.*

H_0^{coll} : **FVGQ-style second-order collateral channel.** *If the true DGP transmits financial frictions mainly through second-order risk corrections in the Euler equation, then*

$$\mathbb{E}_t[\hat{\tau}_{x,t} \mid \Delta\sigma_{e,t}^2] = \kappa_{x,e}\Delta\sigma_{e,t}^2 + o(\Delta\sigma_{e,t}^2),$$

where $\kappa_{x,e}$ is given by (Kappa). In the baseline FVGQ collateral mechanism, the collateral and aggregation components push $\kappa_{x,e}$ toward a positive sign. The after-tax labor wedge does not have a signed first-order negative term driven by the financing shock.

H_1^{wc} : **wage-bill working-capital channel.** *If the true DGP contains a wage-bill working-capital friction and uncertainty or financial tightening raises the external finance spread, then*

$$\mathbb{E}_t[\hat{\Lambda}_{\ell,t} \mid \Delta\Omega_t] = -\lambda_{wc}(\alpha_\xi\rho_{\xi\Omega} + \alpha_\Omega)\Delta\Omega_t + o(\Delta\Omega_t) < 0,$$

provided that $\alpha_\xi\rho_{\xi\Omega} + \alpha_\Omega > 0$. Working capital does not directly create a first-order $\alpha_\xi\hat{\xi}_t$ term in the capital Euler equation.

The testable signatures are summarized below.

Mechanism	Direct condition	Main BCA wedge	Sign pattern
FVGQ collateral / SV risk	Intertemporal Euler, second order	Investment wedge τ_x	$\kappa_{x,e} > 0$ in baseline; $\kappa_{x,A}$ parameter-dependent
Two-agent aggregation	BCA uses C , while investment uses c_E	Investment wedge τ_x plus dispersion term	Financial channel likely reinforces $\tau_x > 0$; macro channel ambiguous
Working capital	Static labor demand, first order	After-tax labor wedge Λ_ℓ	$\hat{\Lambda}_\ell < 0$ when financing spreads rise

Therefore, if the data or simulations show that financial uncertainty mainly lowers $\log(1 - \tau_\ell)$ while the investment-wedge signal is weak, the evidence supports the working-capital channel. If the signal is concentrated in the investment wedge, especially with a positive response of τ_x to financial uncertainty, the pattern is closer to the FVGQ collateral and aggregation channel.

9 Third-Order Projection Extension

Theorem 1 captures only the second-order risk correction $\Phi(\Omega_t)$. This section extends the projection equation to third order in the perturbation parameter, isolating the new objects that enter the BCA wedge under a pruned third-order solution of the FVGQ environment of §3. The main new result is Theorem 3: under standard regularity conditions, the locally bounded solution to the third-order projection equation combines a refined linear coefficient on $\Delta\Omega_t$, a state-dependent cross term in $(\hat{z}_t, \Delta\Omega_t)$, and an explicit asymmetry component that is generated by the exponential structure of the conditional variance.

9.1 Why the Second-Order Projection Is Insufficient for FVGQ

Two observations make a third-order extension necessary in the FVGQ setting.

The FVGQ benchmark uses pruned third order. FVGQ Section 6 solves the model with a pruned third-order perturbation around the deterministic SS. The reason is structural: a second-order policy function contains no terms that involve only the uncertainty shocks $u_{j,t}$. Such terms first appear at third order, in cross-products of the form $\sigma_{j,t-1}u_{j,t}$ and in the constant ν_j^2 that scales the vol-of-vol correction. Without those third-order terms, the impact response of every endogenous variable to a $u_{j,t}$ innovation is identically zero.

Theorem 1 cannot recover impact GIRFs. The Theorem 1 projection coefficient κ_x measures how $\hat{\tau}_{x,t}$ moves with $\Delta\Omega_t$, that is, with the *state* of the conditional variance. It does not measure how $\hat{\tau}_{x,t}$ moves with the *innovation* $u_{j,t}$ that perturbs $\sigma_{j,t}$ at impact. Mechanically, in second order the law of motion of the BCA wedge satisfies

$$\hat{\tau}_{x,t} - \hat{\tau}_{x,t-1} = \kappa_x \rho_{\sigma_j} \Delta\sigma_{j,t-1}^2 + \kappa_x (\sigma_{j,t}^2 - \rho_{\sigma_j} \sigma_{j,t-1}^2) + (\text{not } u_{j,t}).$$

Even though the change $\sigma_{j,t}^2 - \rho_{\sigma_j} \sigma_{j,t-1}^2$ is driven by $u_{j,t}$, that mapping is *linear* only at third order in the perturbation. At second order, $\sigma_{j,t}^2$ is treated as exogenous state without internal dynamics linked to $u_{j,t}$. The local-projection regression $\hat{\tau}_{x,t} \sim u_{j,t}$ is therefore not identified by Theorem 1 alone.

Three new objects. A correctly specified third-order projection introduces the following three new objects beyond the second-order $\Phi(\Omega_t)$:

- $\Psi(\Omega_t)$: pure third-order risk correction, capturing the third cumulant of the log Euler residual conditional on Ω_t .
- $\Gamma(\hat{z}_t, \Omega_t)$: a level-volatility cross term, capturing how the conditional Hessian depends on the level state $\hat{z}_t$.
- Asymmetry contribution: a sign-dependent piece generated by the exponential parametrization $\sigma_{j,t}^2 = \exp(2\sigma_{j,t})$ of the conditional variance.

9.2 Third-Order Expansion of the True Euler Equation

Throughout this section we maintain the FVGQ Euler equation derived in §5:

$$\lambda_t[1 - \theta_t(1 - \delta)] = \beta \mathbb{E}_t G_{t+1}, \quad G_{t+1} \equiv \lambda_{t+1} [R_{t+1}^k + 1 - \delta - \bar{R} \theta_t(1 - \delta)]. \quad (\text{E-Euler-G})$$

The right-hand side is now a single conditional expectation, which makes the log expansion clean. (Note that $R_t = \bar{R}$ at the deterministic SS by the representative-bond pricing condition; we treat R_t

as predetermined when computing $\mathbb{E}_t$.) Taking log on both sides and using the cumulant-generating-function identity

$$\log \mathbb{E}_t[Y] = \mathbb{E}_t[\log Y] + \frac{1}{2} \text{Var}_t(\log Y) + \frac{1}{6} \kappa_3(\log Y \mid \mathcal{F}_t) + \frac{1}{24} \kappa_4(\log Y \mid \mathcal{F}_t) + \dots,$$

applied to $Y = G_{t+1}/\bar{G}$, yields

$$0 = \mathbb{E}_t[m_{t+1} + r_{t+1}^k - \hat{\theta}_{\text{ad},t}] + \frac{1}{2} \Phi(\Omega_t) + \frac{1}{6} \Psi(\Omega_t) + \Gamma(\hat{z}_t, \Omega_t) + o(\sigma^3), \quad (\text{DGP-E3})$$

where:

- $m_{t+1} \equiv \log \lambda_{t+1} - \log \lambda_t$ is the log SDF increment;
- $r_{t+1}^k \equiv \log[R_{t+1}^k + 1 - \delta - \bar{R}\theta_t(1 - \delta)]$ is the log adjusted gross return;
- $\hat{\theta}_{\text{ad},t}$ collects the contemporaneous θ_t -dependent terms (predetermined at t);
- $\Phi(\Omega_t) = \text{Var}_t(\log G_{t+1})$ is the second-cumulant contribution introduced in (FVGQ-Phi);
- $\Psi(\Omega_t) = \kappa_3(\log G_{t+1} \mid \mathcal{F}_t)$ is the third-cumulant contribution; and
- $\Gamma(\hat{z}_t, \Omega_t)$ is a residual cross term capturing the state-dependence of Φ and Ψ themselves.

Structure of Ψ under Gaussian innovations

For a vector of innovations $\varepsilon_{t+1} \sim \mathcal{N}(0, I)$, with $\log G_{t+1}$ admitting the second-order expansion

$$\log G_{t+1} - \overline{\log G} = \xi_1(\Omega_t, \hat{z}_t)' \varepsilon_{t+1} + \frac{1}{2} \varepsilon_{t+1}' \Xi_2(\Omega_t, \hat{z}_t) \varepsilon_{t+1} + o(\|\varepsilon_{t+1}\|^2),$$

direct computation of the third cumulant of a Gaussian quadratic form gives

$$\Psi(\Omega_t) = 8 \text{tr}[(\Xi_2 \Sigma_\varepsilon(\Omega_t))^3] + 6 \xi_1' \Sigma_\varepsilon(\Omega_t) \Xi_2 \Sigma_\varepsilon(\Omega_t) \xi_1 + o(\sigma^3), \quad (\Psi\text{-Gen})$$

where $\Sigma_\varepsilon(\Omega_t)$ is the conditional covariance matrix of ε_{t+1} , which is a linear function of the volatility state Ω_t . The two terms in ($\Psi\text{-Gen}$) are the cubic analogues of the precautionary and risk-premium terms in Φ : the first is “vol-of-vol amplification” and the second is a “state-amplified covariance” contribution.

Structure of Γ

If ξ_1 and Ξ_2 depend on $\hat{z}_t$, then Φ and Ψ also depend on $\hat{z}_t$ through the third-order tensor of $\log G$. Locally,

$$\Gamma(\hat{z}_t, \Omega_t) = \hat{z}_t' \mathbf{G}_{z\Omega} \Delta \Omega_t + \hat{z}_t' \mathbf{G}_{zz} \hat{z}_t \cdot \bar{\Phi}' + \xi_1(\bar{\Omega}, \bar{z})' \frac{\partial \Sigma_\varepsilon}{\partial \hat{z}} \bigg|_{\bar{z}} \hat{z}_t \cdot \Delta \Omega_t + o(\sigma^3), \quad (\Gamma\text{-Gen})$$

where $\mathbf{G}_{z\Omega}$ and $\mathbf{G}_{zz}$ are second-derivative tensors of $\log G$ at the deterministic SS. The first term, $\hat{z}_t' \mathbf{G}_{z\Omega} \Delta \Omega_t$, is the most important new object: it generates a state-dependent projection in which the response of $\hat{\tau}_{x,t}$ to a volatility innovation depends on the level state. We will write this term simply as $\Gamma_{z\Omega} \hat{z}_t \Delta \Omega_t$ in the scalar case below.

9.3 Statement and Proof of the Third-Order Projection

Assumption 3 (Joint linear dynamics). The level state $\hat{z}_t$ and the conditional-variance state Ω_t satisfy a joint VAR(1):

$$\begin{pmatrix} \hat{z}_{t+1} \\ \Delta\Omega_{t+1} \end{pmatrix} = \begin{pmatrix} R_z & 0 \\ 0 & R_\Omega \end{pmatrix} \begin{pmatrix} \hat{z}_t \\ \Delta\Omega_t \end{pmatrix} + \begin{pmatrix} \boldsymbol{\eta}_{z,t+1} \\ \boldsymbol{\eta}_{\Omega,t+1} \end{pmatrix},$$

with $\rho(R_z), \rho(R_\Omega) < 1$ and $\mathbb{E}_t[\boldsymbol{\eta}_{z,t+1} \otimes \boldsymbol{\eta}_{\Omega,t+1}] = 0$ (innovations to the level state and to the volatility state are uncorrelated to leading order).

Assumption 4 (Local differentiability of risk corrections). Φ and Ψ are once differentiable near $\bar{\Omega}$:

$$\Phi(\Omega_t) = \Phi(\bar{\Omega}) + \boldsymbol{\phi}' \Delta\Omega_t + o(\|\Delta\Omega_t\|), \quad \Psi(\Omega_t) = \Psi(\bar{\Omega}) + \boldsymbol{\psi}' \Delta\Omega_t + o(\|\Delta\Omega_t\|).$$

The cross term has the bilinear local form

$$\Gamma(\hat{z}_t, \Omega_t) = \bar{\Gamma} + \bar{\Gamma}'_z \hat{z}_t + \bar{\Gamma}'_\Omega \Delta\Omega_t + \hat{z}'_t \boldsymbol{\Gamma}_{z\Omega} \Delta\Omega_t + o(\sigma^3).$$

Theorem 3 (Third-Order Wedge Projection). *Let Assumptions 3 and 4 hold, and let $|\omega_\tau| < 1$. Then the locally bounded solution to the third-order projection equation*

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t) + \frac{1}{6} \Psi(\Omega_t) + \Gamma(\hat{z}_t, \Omega_t)$$

admits the affine-bilinear form

$$\boxed{\hat{\tau}_{x,t} = \bar{\tau}_x + \boldsymbol{\kappa}_x^{(2)'} \Delta\Omega_t + \boldsymbol{\kappa}_x^{(3,\text{vov})'} \Delta\Omega_t + \bar{\boldsymbol{\kappa}}_z' \hat{z}_t + \hat{z}'_t \boldsymbol{\kappa}_x^{(3,\text{cross})} \Delta\Omega_t + o(\sigma^3)}, \quad (\text{T2})$$

where

$$\bar{\tau}_x = \frac{\Phi(\bar{\Omega})/2 + \Psi(\bar{\Omega})/6 + \bar{\Gamma}}{1 - \omega_\tau}, \quad (\text{T2a})$$

$$\boldsymbol{\kappa}_x^{(2)} = \frac{1}{2} (I - \omega_\tau R_\Omega)^{-1} \boldsymbol{\phi}, \quad (\text{T2b})$$

$$\boldsymbol{\kappa}_x^{(3,\text{vov})} = \frac{1}{6} (I - \omega_\tau R_\Omega)^{-1} \boldsymbol{\psi} + (I - \omega_\tau R_\Omega)^{-1} \bar{\Gamma}_\Omega, \quad (\text{T2c})$$

$$\bar{\boldsymbol{\kappa}}_z = (I - \omega_\tau R_z)^{-1} \bar{\Gamma}_z, \quad (\text{T2d})$$

$$\boldsymbol{\kappa}_x^{(3,\text{cross})} = (I_z \otimes I_\Omega - \omega_\tau (R_z \otimes R_\Omega))^{-1} \boldsymbol{\Gamma}_{z\Omega}. \quad (\text{T2e})$$

Proof. Forward iterating the projection equation as in Theorem 1,

$$\hat{\tau}_{x,t} = \sum_{j=0}^{\infty} \omega_\tau^j \mathbb{E}_t \left[\frac{1}{2} \Phi(\Omega_{t+j}) + \frac{1}{6} \Psi(\Omega_{t+j}) + \Gamma(\hat{z}_{t+j}, \Omega_{t+j}) \right].$$

Substitute the local approximations in Assumption 4 and use Assumption 3 to compute conditional means:

$$\mathbb{E}_t[\Delta\Omega_{t+j}] = R_\Omega^j \Delta\Omega_t, \quad \mathbb{E}_t[\hat{z}_{t+j}] = R_z^j \hat{z}_t, \quad \mathbb{E}_t[\hat{z}_{t+j} \otimes \Delta\Omega_{t+j}] = (R_z^j \otimes R_\Omega^j)(\hat{z}_t \otimes \Delta\Omega_t),$$

where the third equality uses the uncorrelated-innovation condition. Sum the geometric series term by term:

$$\sum_{j=0}^{\infty} \omega_\tau^j R_\Omega^j = (I - \omega_\tau R_\Omega)^{-1},$$

$$\sum_{j=0}^{\infty} \omega_{\tau}^j R_z^j = (I - \omega_{\tau} R_z)^{-1},$$

$$\sum_{j=0}^{\infty} \omega_{\tau}^j (R_z^j \otimes R_{\Omega}^j) = (I_z \otimes I_{\Omega} - \omega_{\tau} (R_z \otimes R_{\Omega}))^{-1}.$$

The constants $\Phi(\bar{\Omega})/2 + \Psi(\bar{\Omega})/6 + \bar{\Gamma}$ aggregate into $\bar{\tau}_x = (\Phi(\bar{\Omega})/2 + \Psi(\bar{\Omega})/6 + \bar{\Gamma})/(1 - \omega_{\tau})$. The remaining identifications yield (T2a)–(T2e). $\square$

Scalar special case. Suppose $\hat{z}_t$ and Ω_t are scalar AR(1) processes with persistences $\rho_z, \rho_{\Omega} \in [0, 1)$ and that the cross-derivative is the scalar $\Gamma_{z\Omega}$. Then (T2) simplifies to

$$\hat{\tau}_{x,t} = \bar{\tau}_x + \kappa_x^{(2)} \Delta \Omega_t + \kappa_x^{(3,\text{vov})} \Delta \Omega_t + \bar{\kappa}_z \hat{z}_t + \kappa_x^{(3,\text{cross})} \hat{z}_t \Delta \Omega_t + o(\sigma^3), \quad (\text{T2s})$$

with $\kappa_x^{(2)} = \phi/[2(1 - \omega_{\tau}\rho_{\Omega})]$, $\kappa_x^{(3,\text{vov})} = \psi/[6(1 - \omega_{\tau}\rho_{\Omega})] + \bar{\Gamma}_{\Omega}/(1 - \omega_{\tau}\rho_{\Omega})$, $\bar{\kappa}_z = \bar{\Gamma}_z/(1 - \omega_{\tau}\rho_z)$, and

$$\kappa_x^{(3,\text{cross})} = \frac{\Gamma_{z\Omega}}{1 - \omega_{\tau}\rho_z\rho_{\Omega}}.$$

Compared to Theorem 1, the persistence factor in $\kappa_x^{(3,\text{cross})}$ is $1 - \omega_{\tau}\rho_z\rho_{\Omega}$ rather than $1 - \omega_{\tau}\rho_{\Omega}$. With $\omega_{\tau} = 0.973$, $\rho_z \approx 0.95$ (the standard FVGQ level persistence), and $\rho_{\Omega} = 0.75$, this gives

$$\frac{1}{1 - \omega_{\tau}\rho_{\Omega}} \simeq 3.70, \quad \frac{1}{1 - \omega_{\tau}\rho_z\rho_{\Omega}} \simeq 2.50.$$

The cross-term coefficient is amplified less than the linear coefficient, but both are nontrivial.

9.4 FVGQ-Specific Third-Order Decomposition

We now apply Theorem 3 to the FVGQ Euler equation. Continuing the decomposition (FVGQ-Phi) into precautionary, risk-premium, collateral, and aggregation components, the third-cumulant correction has the analogous decomposition

$$\Psi^{\text{FVGQ}}(\Omega_t) = \Psi^{\text{prec}}(\Omega_t) + \Psi^{\text{rp}}(\Omega_t) + \Psi^{\text{coll}}(\Omega_t) + \Psi^{\text{agg}}(\Omega_t). \quad (\text{FVGQ-Psi})$$

Precautionary cubic (Ψ^{prec})

Using $\hat{\lambda}_{t+1} = -\gamma\hat{c}_{t+1} - \frac{1}{2}\gamma(\gamma+1)\text{Var}_t(\hat{c}_{t+1}/\sigma) + o(\sigma^2)$ from the second-order expansion of $u'(c)$ under CRRA, the third-cumulant analogue of Φ^{prec} is

$$\Psi^{\text{prec}}(\Omega_t) = -\gamma^3[1 - \bar{\theta}(1 - \delta)] \mathbb{E}_t[\hat{c}_{t+1}^3] + 3\gamma^2(\gamma+1)[1 - \bar{\theta}(1 - \delta)] \mathbb{E}_t[\hat{c}_{t+1} \text{Var}_t(\hat{c}_{t+2} \mid \mathcal{F}_{t+1})]. \quad (\text{Psi-Prec})$$

The first term vanishes for Gaussian innovations of $\hat{c}_{t+1}$ at first order in the perturbation parameter, but it is nonzero at the relevant third order because $\hat{c}_{t+1}$ contains second-order quadratic terms that generate non-vanishing third moments. The second term is the $\mathbb{E}[\hat{c} \cdot \text{Var}(\hat{c})]$ -type correction familiar from the recursive-utility literature.

Risk-premium cubic (Ψ^{rp})

$$\Psi^{\text{rp}}(\Omega_t) = 3\gamma\beta(\bar{R}^k + 1 - \delta) \mathbb{E}_t[\hat{c}_{t+1}^2 \hat{r}_{t+1}^k] - 3\gamma^2\beta(\bar{R}^k + 1 - \delta) \mathbb{E}_t[\hat{c}_{t+1}(\hat{r}_{t+1}^k)^2] + \dots \quad (\text{Psi-RP})$$

Both terms involve mixed moments of consumption and capital returns. They generalize the second-order Σ_{cR} to higher-order coskewness.

Collateral cubic (Ψ^{coll})

This is the FVGQ-specific term, the cubic analogue of the second-order collateral correction $\beta\bar{R}\bar{\theta}(1-\delta)\nu_\theta\sigma_{e,t}^2$. Decompose the third-order entrepreneur log marginal utility as

$$\log \lambda_t = (\text{linear}) + \frac{1}{2}\nu_\theta\sigma_{e,t}^2 + \frac{1}{6}\nu_\theta^{(3)}\sigma_{e,t}^3 + \nu_\theta^{(\text{cross})}\hat{z}_t\sigma_{e,t}^2 + o(\sigma^3),$$

where:

- $\nu_\theta^{(3)} \equiv \partial^3 \mathbb{E}_t \log \lambda_{t+1} / \partial \sigma_{e,t}^3$ is the third-order collateral sensitivity (vol-of-vol response of marginal utility);
- $\nu_\theta^{(\text{cross})} \equiv \partial^3 \mathbb{E}_t \log \lambda_{t+1} / (\partial \hat{z}_t \partial \sigma_{e,t}^2)$ is the cross sensitivity (level-state-dependent collateral response).

Substituting into the Euler equation yields

$$\Psi^{\text{coll}}(\Omega_t) = \beta\bar{R}\bar{\theta}(1-\delta)\nu_\theta^{(3)}\sigma_{e,t}^3, \quad (\text{Psi-Coll})$$

$$\Gamma^{\text{coll}}(\hat{z}_t, \Omega_t) = \beta\bar{R}\bar{\theta}(1-\delta)\nu_\theta^{(\text{cross})}\hat{z}_t\sigma_{e,t}^2. \quad (\text{Gam-Coll})$$

Aggregation cubic (Ψ^{agg})

Recall from §6 that the aggregation component of Φ is $\gamma_{SWSE}[\gamma \text{Var}_t(\delta c_{t+1}) - \text{Cov}_t(\delta c_{t+1}, \hat{r}_{t+1}^k)]$. The third-order analogue introduces the conditional skewness of consumption dispersion and its co-skewness with $\hat{r}_{t+1}^k$:

$$\Psi^{\text{agg}}(\Omega_t) = \gamma_{SWSE} \left[\gamma^2 \mathbb{E}_t[\delta c_{t+1}^3] - 3\gamma \mathbb{E}_t[\delta c_{t+1}^2 \hat{r}_{t+1}^k] + \mathbb{E}_t[\delta c_{t+1}(\hat{r}_{t+1}^k)^2] \right]. \quad (\text{Psi-Agg})$$

Under independent Gaussian innovations to the policy functions of δc and $\hat{r}^k$, all of these third moments are zero at the second-order policy function but become non-zero at third order via the same mechanism as in Ψ^{prec} .

9.5 Reconciling the Monte Carlo Puzzle of $\nu_\theta < 0$

The third-order extension provides a structural explanation for the negative Monte Carlo estimate of ν_θ reported in §10.4. The diagnostic regresses

$$\mathbb{E}[\log \lambda_t \mid \sigma_{e,0} = \bar{\sigma}_e + \delta] \text{ on } \sigma_e^2 = e^{2(\bar{\sigma}_e + \delta)},$$

across an 11-point grid spanning $\bar{\sigma}_e \pm 2\sigma(\sigma_{e,t})$. With $\sigma(\sigma_{e,t}) = 1.05$, the grid extends to $\delta = \pm 2.10$, which corresponds to σ_e^2 ranging over almost three orders of magnitude.

Under the third-order expansion above,

$$\log \lambda = \text{const} + \frac{1}{2}\nu_\theta\sigma_e^2 + \frac{1}{6}\nu_\theta^{(3)}\sigma_e^3 + \frac{1}{24}\nu_\theta^{(4)}\sigma_e^4 + \dots,$$

so a regression on σ_e^2 alone over a wide grid mixes the second-order coefficient ν_θ with the third- and fourth-order coefficients $\nu_\theta^{(3)}, \nu_\theta^{(4)}$. If $\nu_\theta^{(3)} < 0$ and the grid extends far enough that σ_e^3 has a comparable contribution, the OLS slope on σ_e^2 is biased downward. The numerical observation in §10.4 that the central three grid points give a positive slope (consistent with small positive ν_θ) while the wide grid gives a negative slope (-4.93×10^{-2}) is precisely the signature of higher-order contamination. A clean estimate of ν_θ requires the central-difference window to be narrow enough that $|\nu_\theta^{(3)}\sigma_e/(3\nu_\theta)| \ll 1$.

9.6 Implications and Three Testable Predictions

Asymmetric impulse responses

The exponential parametrization $\sigma_{e,t}^2 = e^{2\sigma_{e,t}}$ implies that a positive uncertainty innovation $u_{e,t} = +\delta$ raises σ_e^2 more than a negative innovation $u_{e,t} = -\delta$ lowers it:

$$\sigma_e^2(+\delta) - \bar{\sigma}_e^2 = \bar{\sigma}_e^2(e^{2\nu_e\delta} - 1) > \bar{\sigma}_e^2(1 - e^{-2\nu_e\delta}) = \bar{\sigma}_e^2 - \sigma_e^2(-\delta).$$

Combined with the linear $\kappa_x^{(2)}$ coefficient, the on-impact wedge response satisfies

$$\hat{\tau}_{x,t}(+\delta) + \hat{\tau}_{x,t}(-\delta) \simeq 2\nu_e^2\delta^2\bar{\sigma}_e^2\kappa_x^{(2)} + O(\delta^3).$$

A positive $\kappa_x^{(2)}$ therefore predicts that the average of the responses to $\pm\delta$ uncertainty shocks is strictly positive at impact. This sum can be recovered from a sign-conditional GIRF or local projection.

State-dependent IRFs

The cross term $\kappa_x^{(3,\text{cross})}\hat{z}_t\Delta\Omega_t$ generates state-dependence in the impulse response. Specifically, the response of $\hat{\tau}_{x,t}$ to a unit $u_{e,t}$ shock is approximately

$$\frac{\partial\hat{\tau}_{x,t}}{\partial u_{e,t}} \simeq 2\nu_e\bar{\sigma}_e^2[\kappa_x^{(2)} + \kappa_x^{(3,\text{cross})}\hat{z}_t] + O(\sigma^2).$$

With $\kappa_x^{(3,\text{cross})} > 0$ (as expected when $\nu_\theta^{(\text{cross})} > 0$, which arises naturally if a collateral-tightening regime amplifies the cross-channel), the wedge response should be larger when the economy is already in a collateral-tightening regime ($\hat{z}_t > 0$). This is testable by interacting the LP regressor $u_{e,t}$ with a measure of collateral conditions at t .

Vol-of-vol amplification persistence

In the second-order projection (Theorem 1), the persistence factor on the wedge response is $1/(1 - \omega_\tau\rho_\Omega) \simeq 3.70$. In the third-order projection, the cross-term factor is $1/(1 - \omega_\tau\rho_z\rho_\Omega) \simeq 2.50$, but the pure vol-of-vol contribution within $\kappa_x^{(3,\text{vol})}$ inherits the second-order persistence. A misspecified second-order regression that omits the cross term and absorbs vol-of-vol into the $\kappa_x^{(2)}$ slope therefore biases the estimated persistence toward the longer horizon. This is testable by comparing horizon-by-horizon LP coefficients with and without an explicit $\hat{z}_t \cdot u_{e,t}$ interaction.

10 Numerical Validation from the FVGQ Workspace

The numerical validation uses pruned third-order policy matrices from the FVGQ preliminary workspace. This section should be read as a diagnostic check of the closed-form derivation, not as a final calibration exercise.

10.1 Extracted Coefficients

The extracted coefficients are listed below. The consumption variable is **cspt**, which is composite per-capita consumption. It is not identical to entrepreneur consumption c_E .

Coefficient	Value
$a_{cc,A}$	4.844×10^{-1}
$a_{cc,e}$	4.115×10^{-4}
$a_{cR,A}$	1.886×10^{-2}
$a_{cR,e}$	-1.786×10^{-4}
ν_θ	0 at the first/second-order extraction

The strongest sanity check is the stochastic-volatility loading:

$$\frac{\text{ghxu}[c, \text{sigAt}, e_A]}{\text{ghu}[c, e_A]} = 0.7500 = \rho_{\sigma_A}.$$

This confirms that the channel by which the volatility state changes the one-step-ahead conditional variance is present in the Dynare policy function in exactly the way required by the closed-form derivation.

10.2 Canonical Calibration Used in the Validation

The validation uses the actual calibration in `fvgq2020_solveonly.mod`, not the earlier draft values $\rho_\sigma = 0.95$ and $\bar{\theta} = 0.20$.

Parameter	Value
γ	2.0
β	0.994
δ	0.01625
$\bar{\theta}$	0.0954
$\bar{R}^k$	0.0273
$\bar{R}$	$1/\beta \simeq 1.0060$
$\rho_{\sigma_A}, \rho_{\sigma_e}$	0.75
ω_τ	$\simeq 0.973$
$1 - \omega_\tau \rho_\sigma$	$\simeq 0.270$

The amplification factor is therefore about $1/0.270 \simeq 3.7$, not the roughly 15-fold amplification implied by the earlier, incorrect $\rho_\sigma = 0.95$ value.

10.3 Predicted κ_x and LP Comparison

Without the Monte Carlo correction for ν_θ and without the full two-agent aggregation correction, the extracted coefficients imply

Quantity	Value
ϕ_A	$+8.40 \times 10^{-1}$
ϕ_e	$+1.10 \times 10^{-3}$
$\kappa_{x,A}$	+1.554
$\kappa_{x,e}$	$+2.04 \times 10^{-3}$

Converted into the impact response to a one-unit volatility shock, these figures imply

$$u_A \rightarrow \hat{\tau}_x : +1.06 \times 10^{-4}, \quad u_e \rightarrow \hat{\tau}_x : +1.93 \times 10^{-7}.$$

The comparison with preliminary local-projection peaks is

Channel	Closed-form sign	LP peak sign	Magnitude note
$u_A \rightarrow \tau_x$	Positive	Negative at $h = 40$	Sign mismatch at the peak
$u_e \rightarrow \tau_x$	Positive	Positive at $h = 32$	Correct sign, too small

This mismatch should not be read as a failure of the projection theorem. The closed form is an impact and local projection object, whereas the reported LP number is a long-horizon peak. The comparison should include $h = 0$, the zero crossing, and the peak.

10.4 Monte Carlo Check for ν_θ

A pruned third-order Monte Carlo check was used to extract the collateral-risk term that is not visible in the first/second-order matrix extraction. The simulation starts from the stochastic steady state, overwrites the initial `siget`, and calls the Dynare decision rule through `simult_`. The grid has 11 points over

$$\text{sige} \pm 2\text{std}(\text{siget}), \quad \text{sige} = -4.7975, \quad \text{std}(\text{siget}) = 1.0479,$$

with 1000 simulations per grid point and $T = 200$.

Regressing the first-period cross-sectional Monte Carlo moments on $\sigma_e^2 = \exp(2 \text{sige})$ gives

Quantity	Global-grid slope
$\nu_\theta = \partial \mathbb{E}[\log \lambda] / \partial \sigma_e^2$	-4.933×10^{-2}
$a_{cc,e}^{MC}$	1.635×10^{-4}
$a_{cR,e}^{MC}$	-1.619×10^{-4}

The collateral coefficient is

$$\beta \bar{R} \bar{\theta} (1 - \delta) = 0.093849,$$

so the implied collateral correction per unit of σ_e^2 is

$$0.093849 \nu_\theta = -4.629 \times 10^{-3}.$$

For a one-unit `ue` shock, $\Delta \sigma_e^2 = 9.44 \times 10^{-5}$, so the impact correction is only -4.370×10^{-7} . Adding it to the original $\kappa_{x,e} = 2.04 \times 10^{-3}$ gives

$$\kappa_{x,e}^{\text{revised}} = 2.03956 \times 10^{-3}.$$

Thus the current Monte Carlo estimate of ν_θ is small, negative, and does not supply the missing positive magnitude originally expected from the collateral-risk term.

This estimate should be interpreted as a diagnostic rather than as the final local derivative. The global 11-point regression gives $a_{cc,e}^{MC} = 1.635 \times 10^{-4}$, only 39.7 percent of the policy-function coefficient 4.115×10^{-4} . By contrast, the central three grid points give 4.347×10^{-4} , only 5.6 percent above the policy-function value. The local perturbation check therefore passes near the steady state, while the wide grid is affected by nonlinearities.

10.5 Two-Agent Aggregation Correction

The full aggregation correction was computed from the separate consumption objects `ce` and `cw`. The Dynare policy-function rows were aligned using `order_var`; the first-order aggregation identity

$$\text{cspt} = s_E \text{ce} + s_W \text{cw}$$

then passes as a check. The steady-state consumption shares are

$$s_W = 0.97249, \quad s_E = 0.02751.$$

The extracted one-step innovation loadings are

Channel	ψ_E	ψ_W	χ	r
A	-9.852×10^{-4}	5.812×10^{-3}	-8.410×10^{-1}	2.710×10^{-2}
e	-5.266×10^{-3}	3.477×10^{-4}	-5.892×10^{-1}	-8.806×10^{-3}

These imply

Channel	$\partial \text{Var}(\delta c)/\partial \sigma^2$	$\partial \text{Cov}(\delta c, r^k)/\partial \sigma^2$	ϕ^{agg}
A	7.073×10^{-1}	-2.279×10^{-2}	7.692×10^{-2}
e	3.472×10^{-1}	5.189×10^{-3}	3.688×10^{-2}

Using the same normalization as the rest of the report, $\kappa = \phi/[2(1 - \omega_\tau \rho_\sigma)]$ and $2(1 - \omega_\tau \rho_\sigma) = 0.5405$. Adding the aggregation correction gives

Channel	ϕ original	ϕ updated	κ updated	LP peak
A	8.399×10^{-1}	9.168×10^{-1}	1.696	-1.642×10^{-3}
e	1.105×10^{-3}	3.799×10^{-2}	7.028×10^{-2}	1.886×10^{-3}

The aggregation correction is positive for both channels. It therefore cannot explain the negative LP peak for $u_A \rightarrow \tau_x$; after aggregation, $\kappa_{x,A}$ remains positive and moves further away from the LP peak sign. For u_e , aggregation substantially raises the predicted coefficient, but it does not by itself close the full magnitude gap once the comparison is made in the impact units of the volatility shock.

11 Research Implications and Next Steps

The derivation turns the Year 2 BCA methodology from an informal set of competing hypotheses into a theorem-based framework with testable wedge signatures. The next steps are:

1. **Fix the numerical comparison.** Extract the LP impulse response at $h = 0$, at the zero crossing, and at the peak, rather than comparing only the peak. The current numerical comparison mixes impact closed-form objects with long-horizon LP peaks.
2. **Refine the ν_θ derivative.** The broad-grid Monte Carlo estimate is negative and small, but the central three grid points pass the local perturbation sanity check. The next check should use a narrower grid or a common-random-number central difference around the stochastic steady state.
3. **Reconcile coefficient units.** The aggregation correction substantially raises $\kappa_{x,e}$, but direct comparison to LP peaks depends on whether the reported object is a coefficient per unit σ^2 , an impact response to a unit volatility innovation, or a long-horizon projection peak.
4. **Implement the working-capital comparison model.** Add wage-bill financing to the FVGQ block and verify whether the sign theorem $\hat{\Lambda}_{\ell,t} < 0$ holds in simulation.

5. **Write the formal appendix.** Complete the second-order expansion of (E-Euler), the aggregation wedge derivation, and the working-capital labor projection, with a consistent Φ normalization throughout.
6. **Test the three third-order predictions.** The third-order projection (Theorem 3) yields three new identification handles. (i) Asymmetric IRFs: estimate sign-conditional GIRFs from the FVGQ pruned solution and check whether the response to $u_{e,t} = +\delta$ exceeds the response to $u_{e,t} = -\delta$ in absolute value, and whether the average of the two responses has the predicted sign of $\kappa_x^{(2)}$. (ii) State-dependent IRFs: interact the LP regressor $u_{e,t}$ with a measure of $\hat{z}_t$ (e.g. the contemporaneous level of the θ_t shock or the HP-detrended q_t) and verify whether the cross coefficient is positive. (iii) Vol-of-vol amplification: extract the cubic policy-function coefficient $\nu_\theta^{(3)}$ from the third-order Dynare output and confirm whether it reconciles the wide-grid Monte Carlo regression slope.
7. **Sharpen the $\nu_\theta^{(3)}$ extraction.** Implement a narrow-grid central-difference estimator with common random numbers around the stochastic steady state, computing ν_θ , $\nu_\theta^{(3)}$, and $\nu_\theta^{(\text{cross})}$ jointly. This isolates the structural source of the wide-grid bias documented in §10.4.

Remaining issues that do not undermine tractability include pruning corrections to the risk-adjusted steady state, identification with correlated multi-dimensional volatility states, the non-smooth Hessian created by occasionally binding collateral constraints, and the measurement gap between composite consumption and entrepreneur marginal utility.

A Algebraic Details

A.1 Forward Solution of the Projection Equation

Starting from

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t),$$

repeated substitution gives, for any J ,

$$\hat{\tau}_{x,t} = \frac{1}{2} \sum_{j=0}^J \omega_\tau^j \mathbb{E}_t \Phi(\Omega_{t+j}) + \omega_\tau^{J+1} \mathbb{E}_t \hat{\tau}_{x,t+J+1}.$$

If the solution is locally bounded and $|\omega_\tau| < 1$, the last term converges to zero. Substituting $\mathbb{E}_t \Delta \Omega_{t+j} = R_\Omega^j \Delta \Omega_t$ yields Theorem 1.

A.2 Working-Capital Labor Wedge

The working-capital firm condition and household condition are

$$A_t F_{L,t} = w_t (1 + \theta_{wc} q_t), \quad -\frac{U_L}{U_C} = w_t.$$

The BCA labor condition requires a wedge such that

$$w_t = \Lambda_{\ell,t}^{BCA} A_t F_{L,t}.$$

Therefore

$$\Lambda_{\ell,t}^{BCA} = \frac{1}{1 + \theta_{wc} q_t}.$$

Taking logs and expanding around $\bar{q}$ gives

$$\log \Lambda_{\ell,t}^{BCA} = -\log(1 + \theta_{wc} q_t) \simeq -\frac{\theta_{wc}}{1 + \theta_{wc} \bar{q}}(q_t - \bar{q}).$$

This is equation (WC-LIN).

A.3 Two-Agent Consumption Aggregation

Let $c_{i,t} = \bar{c}_i \exp(\hat{c}_{i,t})$. Then

$$\frac{C_t}{\bar{C}} = s_W \exp(\hat{c}_{W,t}) + s_E \exp(\hat{c}_{E,t}).$$

Taking logs and expanding to second order gives

$$\hat{C}_t = s_W \hat{c}_{W,t} + s_E \hat{c}_{E,t} + \frac{1}{2} [s_W \hat{c}_{W,t}^2 + s_E \hat{c}_{E,t}^2 - (s_W \hat{c}_{W,t} + s_E \hat{c}_{E,t})^2] + O(3).$$

The bracketed term simplifies to $s_W s_E (\hat{c}_{E,t} - \hat{c}_{W,t})^2$, which proves (Agg-C).